

Inverse Teleprompter (Rolling Display with voice control)

Aim

To design and implement a Bluetooth Voice-to-Text based LED Matrix Display using an ESP32-C6 microcontroller, HC-05 Bluetooth module, and a MAX7219 LED matrix, where spoken input from a smartphone is converted to text using a Voice-to-Text app and transmitted over Bluetooth to be displayed as scrolling text on the LED matrix.

Components Needed:

S. No	Component	Quantity
1.	ESP32-C6 Dev Board	1
2.	4-in-1 MAX7219 LED Matrix	1
3.	Breadboard	1
4.	Jumper Wires	As needed
5.	USB Type-C Cable	1
6.	Computer with Arduino IDE	1
7.	Bluetooth Module Hc-05	1

Theory:

The project is based on converting speech into text and displaying it wirelessly on an LED matrix. A Voice-to-Text mobile app is used to change spoken words into text, which is then sent to the ESP32-C6 through the HC-05 Bluetooth module. The ESP32 receives this text and drives the MAX7219-based LED matrix using the MD_Parola library to scroll the message across the display. In this way, the system combines voice recognition, Bluetooth communication, and matrix control to create a simple and effective voice-to-text display board.

Circuit Diagram (Connections):

MAX7219 LED Matrix (to ESP32-C6)

- DIN → GPIO23
- CS → GPIO5
- CLK → GPIO18
- VCC → 5V
- GND → GND

HC-05 Bluetooth Module (to ESP32-C6)

- HC-05 TX → ESP32 RXD2 (GPIO6)
- HC-05 RX → ESP32 TXD2 (GPIO7) (use voltage divider to step down 3.3V if needed)
- VCC → 5V
- GND → GND

Procedure:

1. Hardware Setup:

- Connect the MAX7219 module to ESP32-C6 as per pin mapping (DIN → GPIO23, CS → GPIO5, CLK → GPIO18).
- Connect the HC-05 Bluetooth module to ESP32's UART1 (RXD2 = GPIO6, TXD2 = GPIO7).
- Keep the HC-05 EN pin floating (AT mode not required).
- Power the ESP32-C6 and modules using 5V supply through USB.

2. Software Setup:

- Install Arduino IDE on your computer.
- Install required libraries from Library Manager:
 - MD_Parola - (By majicDesigns)
 - MD_MAX72XX - (By majicDesigns)
- Install ESP32 Board Support in Arduino IDE.
- Install Arduino Bluetooth Controller App [link](#)
- Select ESP32C6 Dev Module and choose the correct COM port.

3. Program Upload:

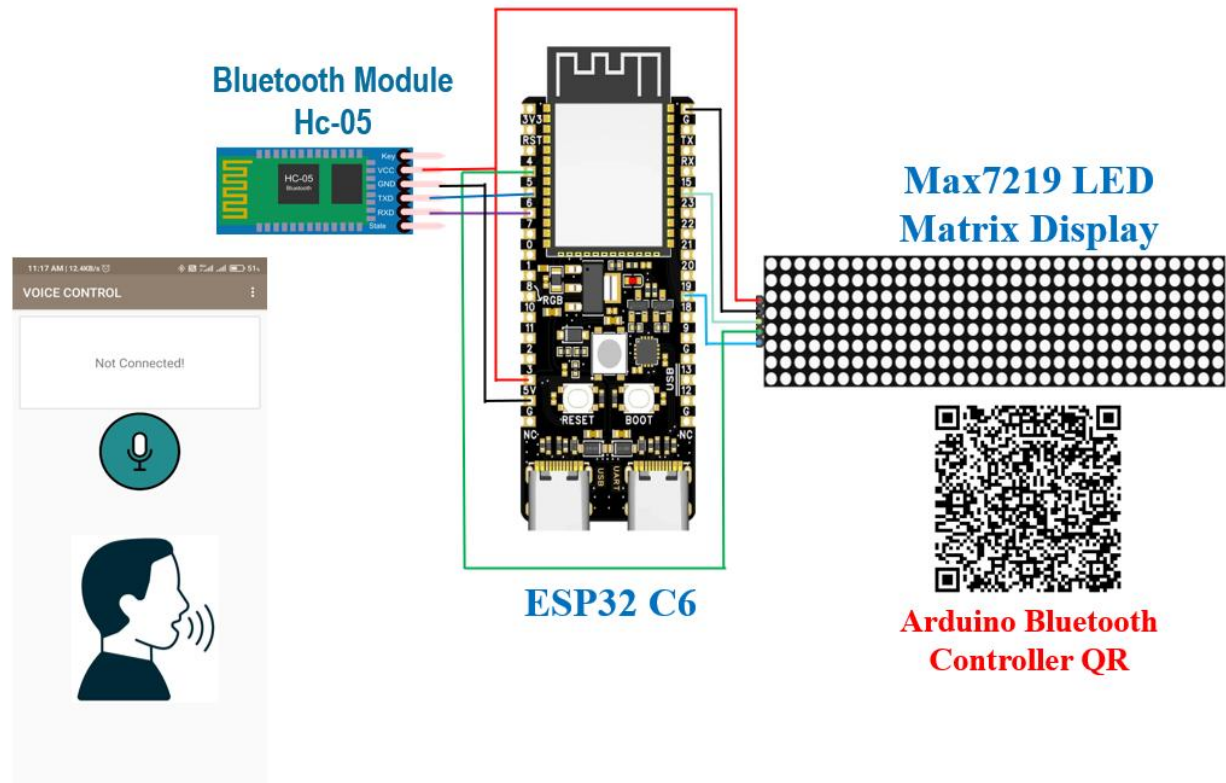
- Copy the provided Arduino sketch into the IDE.
- Verify and upload the code to ESP32-C6.

4. Testing & Operation:

- Pair the HC-05 Bluetooth module with your smartphone (default PIN: 1234 or 0000).
- Open a Voice-to-Text Bluetooth Terminal app on the smartphone.
- Speak into the app → the app converts speech into text and sends it via Bluetooth.
- The ESP32 receives the text and scrolls it on the LED matrix display.

Circuit Diagram:

Bluetooth Voice Based Matrix Display



Software Code:

```
#include <HardwareSerial.h>
#include <MD_Parola.h>
#include <MD_MAX72XX.h>
#include <SPI.h>

// ----- BLUETOOTH CONFIGURATION -----
#define RXD2 6
#define TXD2 7
HardwareSerial BTSerial(1); // Use UART1

// ----- MATRIX DISPLAY CONFIGURATION -----
#define HARDWARE_TYPE MD_MAX72XX::FC16_HW
#define MAX_DEVICES 4
#define DATA_PIN 23
#define CS_PIN 5
```



```
#define CLK_PIN 18

MD_Parola matrix = MD_Parola(HARDWARE_TYPE, DATA_PIN, CLK_PIN,
CS_PIN, MAX_DEVICES);

// ----- VARIABLES -----
String receivedText = ""; // For storing received text

void setup() {
  Serial.begin(115200); // Debug via USB
  BTSerial.begin(9600, SERIAL_8N1, RXD2, TXD2); // Bluetooth

  matrix.begin();
  matrix.setIntensity(5); // Brightness (0–15)
  matrix.displayClear();
  matrix.displayText("READY", PA_CENTER, 100, 0, PA_SCROLL_LEFT,
PA_SCROLL_LEFT);
  Serial.println("Bluetooth Matrix Display Ready");
}

void loop() {
  // Keep display running animations
  if (matrix.displayAnimate()) {
    matrix.displayReset();
  }
  // Check Bluetooth data
  if (BTSerial.available()) {
    receivedText = BTSerial.readStringUntil('\n');
    receivedText.trim();
    Serial.print("Received: ");
    Serial.println(receivedText);
    // Show text on matrix
    matrix.displayClear();
    matrix.displayText(receivedText.c_str(), PA_CENTER, 100, 0,
PA_SCROLL_LEFT, PA_SCROLL_LEFT);
  }
}
```